

## RESEARCH ARTICLE

# Learning From Good Maps: A Dataset and Analysis of Layout Patterns

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## ABSTRACT

The layout of thematic maps is fundamental to cartographic design, serving to effectively utilize map space, construct visual balance, and guide the user's visual attention toward key thematic narratives. Despite the long tradition of cartographic design principles, there remains a need for large-scale, data-driven studies that explicitly extract and analyze interpretable layout patterns for cartographic evaluation. In this study, we construct a large, high-quality dataset of 1625 exemplary maps collected from multiple authoritative and creative sources. Then we develop an automated pipeline for detecting cartographic elements and extracting shape of map area using a fine-tuned YOLOv8-OBB model and the Segment Anything Model (SAM). Based on these detected elements, we establish a set of quantitative metrics for evaluating the design effectiveness of map layouts, focusing on hierarchy, balance, compactness, and alignment. Statistical analysis, including correlation tests and Scheirer-Ray-Hare variance analysis, reveals significant relationships between page layout, map area shape, and layout performance. The results reveal that many widely accepted design conventions, such as the central placement of the map area, a strong tendency toward visually balanced layouts and the typical positioning of titles and legends, are strongly supported by empirical evidence. However, the analysis also exposes notable gaps between conventional practice. For example, although the north arrow is commonly regarded as a standard cartographic element, it appears in only a very small fraction of exemplary maps (1.85%). In addition, the findings uncover substantial flexibility in professional layout strategies, particularly in alignment, where moderate rather than strict geometric alignment predominates. Significant interaction effects further indicate that optimal layout strategies are strongly shape-dependent, highlighting that cartographic design cannot be reduced to a single universal template. Finally, we also demonstrate how the dataset and framework can support future layout exploration, evaluation, and recommendation.

## 1 | Introduction

A map is a composite of multiple elements, including title, map area, inset map, north arrow, scale, and legend, which collectively convey spatial information to the users (Jia et al. 2024). The layout of these elements is crucial for ensuring clarity, readability, and overall effectiveness. A well-organized layout not only enhances comprehension but also guides the user's attention by creating a balanced arrangement that distinguishes

between primary and secondary elements (Wesson 2017). This helps users quickly find key information and make informed decisions (Wei et al. 2017; Dayama et al. 2020; Zeng et al. 2023). Therefore, the importance of a rational and effective map layout cannot be overstated, as it directly impacts the map's utility and the quality of spatial information conveyed.

To support effective map layout, numerous guidelines have been developed, primarily based on human visual perception,

as maps are inherently visual products (MacEachren 2004). These guidelines are mainly generated from psychological principles involving vision perception, such as visual hierarchy, visual balance, and Gestalt principles (Tait 2018). Over the past few decades, these principles have been refined and elaborated by scholars such as Dent et al. (2008), Slocum et al. (2009), and Brewer (2016), becoming foundational standards for map layout design. Despite these guidelines, users often still face challenges in creating layouts that meet their satisfaction. To address these challenges and support the automatic or semi-automatic generation of map layouts, researchers such as Wei et al. (2017) and Dayama et al. (2020) have sought to translate these guidelines into actionable knowledge that can effectively guide the design process. These studies can be categorized into two primary approaches. The first approach focuses on providing users with effective layout knowledge or guidance before or during the design process. This includes integrating exemplary map layout cases as expert knowledge within systems to offer personalized recommendations tailored to users' specific needs (Wei et al. 2017), developing these guidelines into explicit rules for layout (Qin and Li 2017), or incorporating them into optimization algorithms such as Integer Programming to generate satisfactory layouts (Dayama et al. 2020). The second approach involves evaluating the completed layout, assessing factors such as spatial balance (Ma et al. 2013), the importance of structure (Krygier and Wood 2016), and aesthetic considerations (Zhang et al. 2018).

Despite these advancements, designing a successful map layout is still not an easy task. First, while exemplary map layout cases are often used as a knowledge base for layout reuse or recommendations, the limited scope of these cases can be problematic. This limitation arises from the difficulty of automatically recognizing layouts from maps, resulting in systems that may be overly specific and not adaptive to a wide range of map designs. Second, while metrics for evaluating spatial balance and structural importance are available, they often focus only on specific aspects of the map layout. The overall quality of a map layout should be assessed in a more comprehensive manner, considering all relevant guidelines. Furthermore, although these metrics have been developed by experts, they often lack validation against large-scale data, which limits their generalizability. As a result, such evaluations can be somewhat subjective and may not always offer practical guidance for users in real-world design situations. Therefore, a systematic analysis on the map layout quality in more comprehensive datasets is still necessary.

Thanks to the growing collection of well-designed maps available online today, along with the advancement of deep learning algorithms, we now have the opportunity to efficiently recognize and analyze map layouts using object detection techniques (Redmon et al. 2016). This technological progress has already enabled studies like MapLayNet (Yang et al. 2025), which provides a dataset and conducts data-driven analysis of map layouts. While such efforts focus on layout retrieval, our work seeks to focus on identifying patterns—specifically, those emerging from “good maps” that exemplify strong cartographic design. Therefore, if we can define a systematic layout framework and curate high-quality data, we will be able to build upon these technologies to identify these patterns more comprehensively. This, in turn, will offer valuable insights into what constitutes a

good map layout and provide a solid foundation for future automated or semiautomated map design systems.

Building on the above motivation, this paper presents a data-driven framework for discovering effective layout patterns from high-quality thematic maps. Specifically, we first collected over 20,000 relatively high-quality maps from multiple authoritative sources and manually screened them to construct a curated dataset of 1625 exemplary thematic maps. Figure 1 showcases its diversity across multiple dimensions, including element quantity, horizontal versus vertical orientations of map area shapes, and varying page aspect ratios from elongated horizontal to elongated vertical formats. Based on this dataset, we then develop multiple models for extracting map layout elements and the shape of the map area from the dataset using object detection and semantic segmentation techniques. Then using the extracted layouts from these collected maps, we establish a quantitative evaluation framework to assess map layouts from multiple perspectives, followed by comprehensive analysis to figure out what make a good map layout. Finally, we discuss the potential usages that allow users to explore this dataset and receive intelligent layout recommendations tailored to their specific needs.

## 2 | Related Works

### 2.1 | Image Dataset in Map

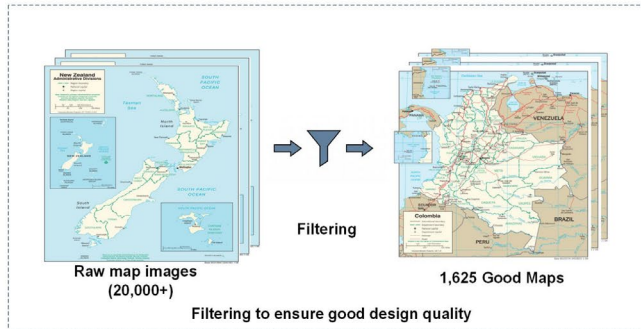
Early research on map image datasets mainly focused on scanned land maps and classification tasks. For example, Mandal et al. (2014) provided a land map image database which included 446 annotated map images across categories such as political, physical, resource, and topographic, supporting studies on text extraction and map-type classification. Following this, Zhou et al. (2018) introduced a dataset of 1812 maps spanning seven categories, which enabled deep convolutional neural networks to classify maps into types like topographic, terrain, or urban scenes. At the generative frontier, the Pix2Pix Maps dataset paired Google Maps street renderings with aerial photos to enable image-to-image translation (Isola et al. 2017), while segmentation-focused datasets from USGS topographic maps (Chiang et al. 2019; Saeedimoghaddam and Stepinski 2020) or historical topographic maps (Kramm et al. 2025) offered annotated road and feature data for deep learning-based digitization of maps. However, these early datasets were largely designed for thematic content recognition rather than the structural or aesthetic aspects of map layouts.

More recent efforts have emphasized broader coverage and fine-grained cartographic structure. The CartoMark dataset expanded to multitask benchmarks including map text annotation, map classification, and style transfer, consolidating maps from diverse sources (Zhou et al. 2024). The ubiMap dataset and its refined subset ubiMap-I annotated cartographic elements across 3500+ maps, enabling structural layout analysis and powering models like MapLayNet for layout retrieval (Yang et al. 2025). Complementarily, the ICDAR 2024 MapText dataset curated 940 high-resolution historical map crops from the David Rumsey Collection with detailed place-name annotations, specifically targeting text detection and recognition (Li et al. 2024).

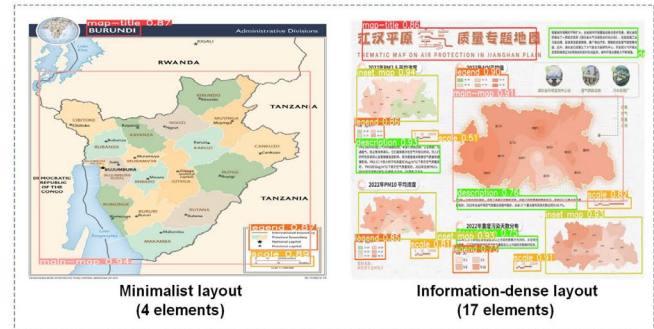
## Comprehensive Showcase of the Thematic Map Layout Dataset

1,625 Curated Map (from 20,000+ raw samples)

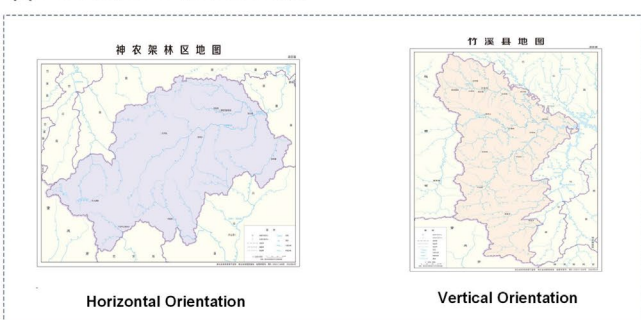
### (a) Good map selection



### (b) Diversity in Element Quantity



### (c) Diversity in Map Area Shape



### (d) Diversity in Page Layout



**FIGURE 1** | Visual examples demonstrating the diversity of map layouts in our dataset. All thumbnails were selected from the final curated dataset of 1625 exemplary thematic maps collected from the four source categories described in Section 3.1. The figure illustrates (a) the curation process for design quality, and variations in (b) element quantity, (c) map area shape, and (d) overall page layout.

Beyond structural annotations, behavioral datasets have also emerged. For instance, He et al. (2023) developed a geospatial image-based eye movement dataset for cartography and GIS, which records gaze behavior as participants interact with different map types. These advances indicate a shift toward datasets that consider maps as structured visual documents, yet they still fall short in directly addressing map design principles and evaluation.

In summary, most existing map image datasets have been designed to support visual recognition tasks such as classification, segmentation, and text detection, with only limited attention to cartographic design and layout. This gap underscores the need for datasets and analytical frameworks that explicitly focus on layout structures, which is also the core motivation of our work.

## 2.2 | Map Layout Design

Early studies on map layout design primarily focused on formulating principles and guidelines for effective composition. Cartographic literature emphasized bottom-up rules such as alignment, balance, and spacing (MacEachren 2004; Dent et al. 2008; Brewer 2016; Slocum et al. 2009), while other works introduced top-down frameworks to classify layouts into categories. For instance, Wu et al. (2012) proposed a taxonomy of layout types based on spatial organization, and Xu et al. (2024) further consolidated these into several representative forms,

highlighting visual balance and compositional harmony as key evaluation dimensions. However, these works remained largely guidance-oriented rather than tool-oriented, offering valuable theory without operational mechanisms for automatic or semi-automatic layout generation.

With the advancement of computing technologies, beyond the guidelines and principles, researchers increasingly sought to quantify layout quality and design computational methods for layout optimization. On the evaluation side, visual balance was modeled via moment-balance mechanics (Ma et al. 2013), while works in cartography and visualization proposed indices for hierarchy and aesthetic assessment or responsive adaptation (e.g., Krygier and Wood 2016; Zeng et al. 2023). On the generation side, studies encoded expert knowledge into actionable rules or optimization procedures: integrating exemplary cases for recommendation (Wei et al. 2017), formalizing legend grouping rules (Qin and Li 2017), and employing integer programming for interactive layout design (Dayama et al. 2020). Together these efforts show a shift from qualitative guidance to operationalizable methods that can evaluate and compose layouts under constraints. Beyond cartography, similar quantitative approaches have been applied to documents, posters, and user interfaces, where design metrics are used to capture principles of hierarchy, alignment, and harmony (O'Donovan et al. 2014; Deka et al. 2017). More recently, learning-based approaches have emerged to capture layout structures from large-scale datasets. Generative models such as LayoutGAN (Li et al. 2019) and LayoutVAE (Jyothi et al. 2019) learn to synthesize layouts

with aesthetic plausibility, while graph-based methods such as LayoutGMN (Patil et al. 2021) model layouts as structured graphs to better represent element relationships. Building on these advances, cartographic layout research has begun to adopt data-driven recognition models. For example, MapLayNet (Yang et al. 2025) proposed a structure-aware graph neural network to represent map layouts as graphs of elements, enabling weakly supervised learning for layout similarity and classification. However, these approaches mainly focus on implicit layout embeddings and downstream tasks, rather than explicitly revealing interpretable cartographic design principles. This highlights the need for a comprehensive framework that combines scalable data-driven analysis with explicit layout knowledge discovery.

In summary, prior work has laid important foundations by defining principles, proposing categorical taxonomies, and introducing quantitative indices for layout evaluation. At the same time, recent computational approaches highlight the potential of data-driven methods to scale analysis and enable automated design. However, existing studies remain fragmented, either focusing on small-scale taxonomies or isolated metrics. This underscores the necessity of our work, which aims to build a large dataset and establish a comprehensive evaluation framework to analyze the design patterns in map layout.

### 3 | Dataset Construction

In this section, we overview the construction of the map dataset for later map layout analysis, which involves three main steps: (1) collecting high-quality map data (Section 3.1), annotating cartographic elements (Section 3.2) and extracting the shape of the map area (Section 3.3).

#### 3.1 | Collection of the Maps

The primary goal of this study is to analyze the characteristics of layouts in exemplary cartographic works. To ensure the structural richness and representativeness of layout designs, the scope of this dataset was strictly limited to thematic maps, whose layouts typically allow greater design flexibility than standardized topographic maps. To achieve this, we first collected more than 20,000 candidate maps from multiple domestic and international sources, specifically including:

- Cartographic Competitions and Exhibitions (CCE): Award-winning maps from the SuperMap Cup, the Xinghu Cup (two widely participated national cartography contests in China), the International Cartographic Conference (ICC) Map Gallery, and the Atlas of Design curated by the North American Cartographic Information Society (NACIS).
- Atlases and Academic Collections (AAC): Authoritative map collections such as the Atlas of Guangdong Province (a highly evaluated and open sourced atlas in China), the China Map Atlas Resource Network (an authoritative online atlas repository in China), and the Harvard Map Collection.

- Official and Institutional Sources (OIS): Maps from Tianditu (China's National Map Service Platform) and the United Nations Cartographic Section are standardized practices in government mapping and international cartographic conventions.
- Visualization Platforms (VP): Design-rich thematic maps from visualization platforms such as Visual Capitalist; they provide aesthetic appeal and effective communication of statistical data.

We then conducted a multistage manual screening process based on visual quality, design completeness, thematic relevance, and suitability for layout analysis, removing duplicated, low-resolution, distorted, or structurally incomplete maps. Each candidate map was independently reviewed by three evaluators, and only maps unanimously approved by all three were included in the final dataset. Through this rigorous dataset construction process, 1625 high-quality exemplary thematic maps were finally retained. Our data collection strategy prioritized maps from the diverse authoritative sources rather than categorizing them by semantic themes, because in cartographic theory the layout structure of a thematic map is more strongly shaped by spatial extent, map morphology, and the management of “negative space” than by thematic topic itself (Dent et al. 2008; Brewer 2016; Slocum et al. 2009). These maps finally serve as reliable references for extracting representative layout patterns and principles. The thumbnails shown in Figure 1 were selected from this final curated dataset to illustrate the diversity of sample sources and layout characteristics.

#### 3.2 | Annotation of the Map Elements

To enable a better understanding of the map layout, we then identify the map elements in a layout. Therefore, we constructed an annotated map layout dataset and trained an object detection model to identify the map elements (Section 3.2.2). The classification of map elements was derived from established cartographic design literature (e.g., Dent et al. 2008; Slocum et al. 2009; Brewer 2016; Dastrup 2020) (Section 3.2.1).

##### 3.2.1 | Element Categories

Different authors have proposed varying definitions and categorizations of map elements, particularly in relation to map layout and design. Classical cartographic texts (e.g., Dent et al. 2008; Slocum et al. 2009) identify a core set of fundamental components, including the map area, title, legend, and scale. Later works refine or expand these categories—for example, Brewer (2016) highlights the role of visual hierarchy and introduces supporting elements such as inset maps, north arrows, or explanatory text blocks where appropriate. Other perspectives emphasize the distinction between essential and optional elements, acknowledging that features like neatlines, graticules, source credits, or metadata appear only in specific contexts (Dastrup 2020). In line with these established discussions, this study adopts and adapts prior definitions to suit its scope. Table 1 summarizes the used types of map elements considered in this study, along with their definitions.

### 3.2.2 | Map Element Detection Model

From the full collected map dataset (1625 maps), 600 samples were selected for manual annotation. These samples were evenly drawn from the four source categories (CCE, AAC, OIS, and VP) to ensure diversity and representativeness. Although we do not perform category-specific layout comparisons in our subsequent analysis, this balanced sampling strategy is crucial. It prevents the deep learning models from overfitting to the specific design templates or spatial constraints of any single publisher, thereby maximizing the structural diversity and representativeness of our annotated subset. Each map was annotated independently by two annotators, following the classification scheme of map elements defined in Section 3.2.1. Specifically, the spatial extent of each cartographic element was annotated using an Oriented Bounding Box (OBB) rather than a complex polygon. Annotations that were consistent between annotators were directly accepted, while disagreements were resolved through discussion to

**TABLE 1** | Definitions of cartographic element types considered in this study.

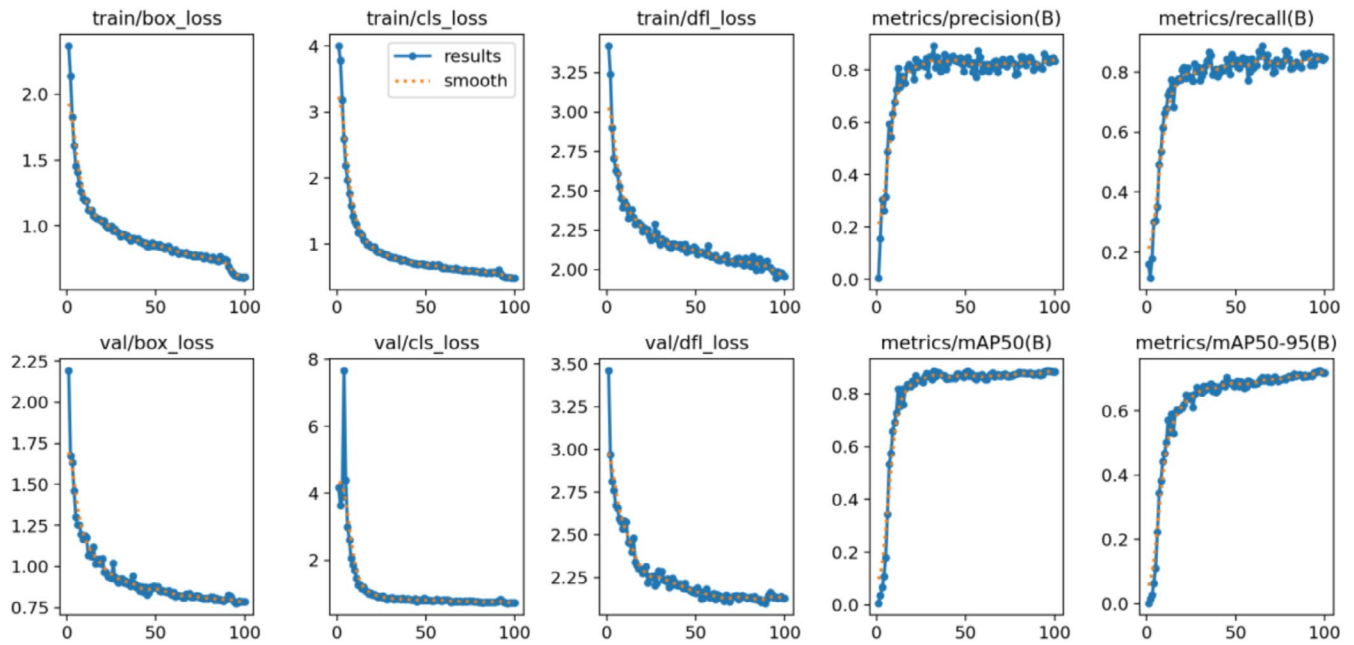
| Element     | Description                                                                                                                                    |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Title       | Textual element that provides the main subject or theme of the map, usually placed in a prominent position to guide the reader's attention.    |
| Map area    | The primary geographic area depicted, serving as the central focus of the layout.                                                              |
| Lengend     | A reference box explaining the meaning of symbols, colors, or patterns used in the map area.                                                   |
| Scale       | A graphical or numerical indicator that expresses the ratio between map distance and real-world distance.                                      |
| Inset map   | A supplementary map embedded within the layout, often used to show location context, enlargements, or global positioning.                      |
| Statistic   | The visualizations of statistical data, such as charts, graphs, or other forms of data representation designed to support the map's narrative. |
| Description | Explanatory or descriptive textual content (excluding the title), used to clarify background, context, or additional information.              |
| North arrow | A directional indicator pointing to geographic north, assisting orientation.                                                                   |
| Picture     | Non-cartographic images (e.g., photos, diagrams) added to enrich the visual narrative of the map.                                              |
| Table       | Structured tabular information included within the layout, typically providing supporting statistical or categorical data.                     |

reach a consensus. The final annotated subset (569 maps) was then used to fine-tune a YOLOv8-OBB model, which had been pretrained on a large-scale object detection dataset (Mitrev and Mirceva 2025). Fine-tuning was carried out with an 80/20 split, using 80% of the annotated maps for training and 20% for testing.

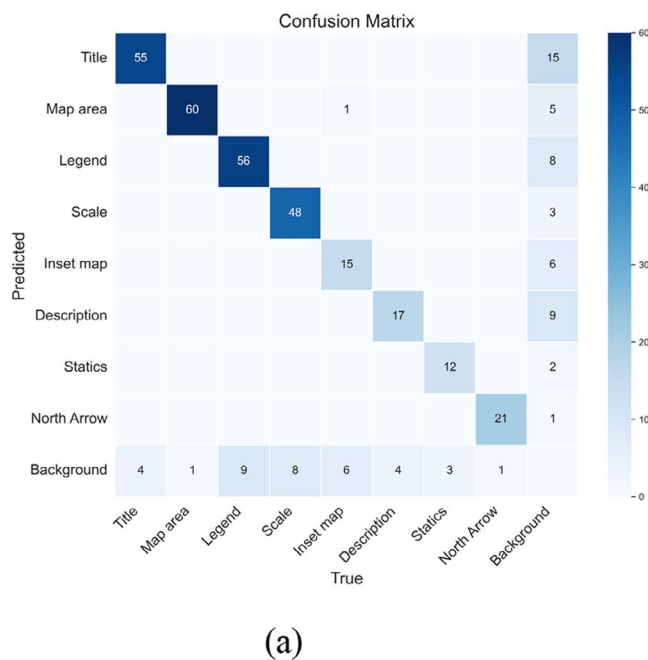
**3.2.2.1 | Training Details.** In the fine-tuning process, the batch size was set to 128, and all input images were resized to  $640 \times 640$  pixels, which can provides a good trade-off between resolution and computational cost. Training was conducted for 100 epochs. For optimization, the optimizer was set to auto, allowing the framework to automatically select the most suitable optimization strategy. The initial learning rate was set to 0.01, with a final learning rate decay factor of 0.01 to ensure gradual convergence during later training stages. In addition, a momentum term (0.937) was introduced to accelerate training and stabilize gradient updates, while a weight decay parameter (0.0005) was employed to improve the model's generalization ability and prevent overfitting.

**3.2.2.2 | Training Loss and Accuracy.** The training loss and accuracy across 100 epochs for both training and validation sets are shown in Figure 2. As shown in the figure, we can observe that all major loss components—including box regression loss (box\_loss), classification loss (cls\_loss), and distribution focal loss (dfl\_loss)—exhibited clear downward trends and eventually converged. This indicates that the model gradually improved both localization and classification accuracy. Importantly, the loss curves of the training and validation sets followed a similar trajectory without significant divergence, suggesting that the model did not suffer from overfitting or underfitting. Meanwhile, the detection metrics—precision, recall, and mean Average Precision (mAP)—showed consistent improvement during training. Both precision and recall rose rapidly in the early epochs, with precision stabilizing around 0.85 and recall around 0.85 in the final stages. The key evaluation indicators also reached satisfactory levels: mAP@0.5 stabilized around 0.88, and mAP@0.5:0.95 stabilized around 0.72. These results demonstrate that the fine-tuned YOLOv8-OBB model has achieved reliable generalization ability for detecting map elements.

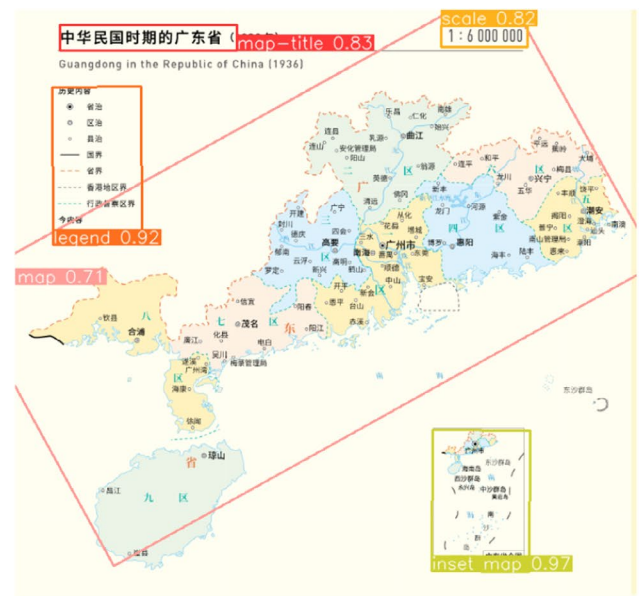
To further evaluate the classification accuracy and misclassification patterns of the fine-tuned model, we generated the confusion matrix on the validation set, as shown in Figure 3a. The rows of the matrix represent the ground truth categories, while the columns represent the predicted categories. The values on the main diagonal indicate the proportion of correctly classified samples (True Positives), where a darker color signifies higher accuracy. As shown in the figure, we can observe that the matrix exhibits a strong diagonal structure, indicating that the model achieves high classification accuracy for the majority of map elements. Furthermore, categories with distinct visual features and spatial significance, such as “Map Area (60/66)”, “Map Legend (56/64)” and “Scale (48/51),” achieved highly accurate detection for core layout elements, highlighting the model's ability to capture their unique characteristics. Example map element detection results are shown in Figure 3b. As shown, the model outputs the detected elements along with their bounding boxes and confidence scores.



**FIGURE 2** | Learning curves of the model over 100 training epochs. The top row displays the training loss metrics (box regression, classification, and distribution focal loss) alongside precision and recall. The bottom row presents the corresponding validation losses and the mean Average Precision (mAP) scores at IoU thresholds of 0.50 and 0.50–0.95. Within each subplot, the solid blue line represents the raw logged results per epoch, while the dotted orange line indicates the smoothed trend.



(a)



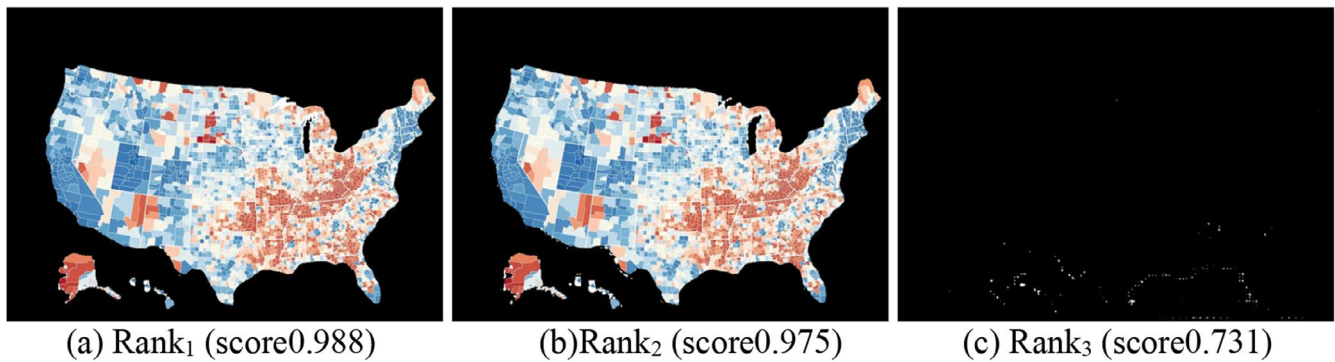
(b)

**FIGURE 3** | Confusion matrix (a) and example detection result (b).

### 3.3 | Map Area Shape Extraction

The map area is the central component of a cartographic layout. Unlike other elements such as titles, legends, and scale bars, which are usually regular in shape and can be adequately represented by bounding boxes, the map area often has highly irregular boundaries and strongly influences whitespace distribution

and the placement of surrounding elements. Therefore, accurately delineating its boundary is essential for analyzing layout structure. Object detection in Section 3.2 provides only an oriented bounding box, which is insufficient to capture the irregular shapes of diverse map areas. Accordingly, based on the detection results, we further refine the map area in two stages: segmentation with the Segment Anything Model (SAM) (Kirillov



**FIGURE 4** | Segmentation results using SAM.

et al. 2023) to obtain precise shape boundaries, followed by post-processing to remove noise and ensure clean representations.

### 3.3.1 | Segmentation with SAM

We adopted SAM, a foundation model for image segmentation that demonstrates strong generalization ability across heterogeneous datasets to obtain the shape boundaries of the map area (Kirillov et al. 2023). But as the model may occasionally generate abnormal or erroneous structures due to variations in image quality or the complexity of the map elements. To address this, we generate multiple candidate segmentation masks using SAM, with the bounding box predicted for the map area as the input prompt. A score-based re-ranking strategy was then applied via the object detection model trained in Section 3.2. As shown in Figure 4, three candidate segmentation masks are initially produced. The mask with the highest confidence score (Rank 1, 0.988) is selected as the final segmentation result, while abnormal candidates, such as the one shown in Figure 4c, are filtered out through this process.

### 3.3.2 | Post-Processing

The raw segmentation masks generated by SAM sometimes contained small artifacts or fragmented regions, especially in layouts with dense surrounding elements. Recent studies have also highlighted that ensuring boundary continuity remains a challenge for foundational models in complex scenes (Ke et al. 2023). To refine these outputs, we applied a sequential post-processing pipeline. First, we implemented noise filtering based on Connected Component Analysis (CCA) (Rosenfeld and Pfaltz 1966) to identify and eliminate disconnected “island” components smaller than a predefined relative area threshold. Specifically, any disconnected component with an area smaller than 2% of the initial bounding box area is classified as noise and eliminated. Subsequently, we applied morphological smoothing (Haralick et al. 1987)—specifically a closing operation followed by opening—to fill internal voids and simplify jagged edges caused by pixel-level prediction errors. An example result is shown in Figure 5. As seen in Figure 5b, small disconnected polygons are present in the raw mask, while the final refined shape of the map area is shown in Figure 5c.

## 4 | Layout Analysis Framework

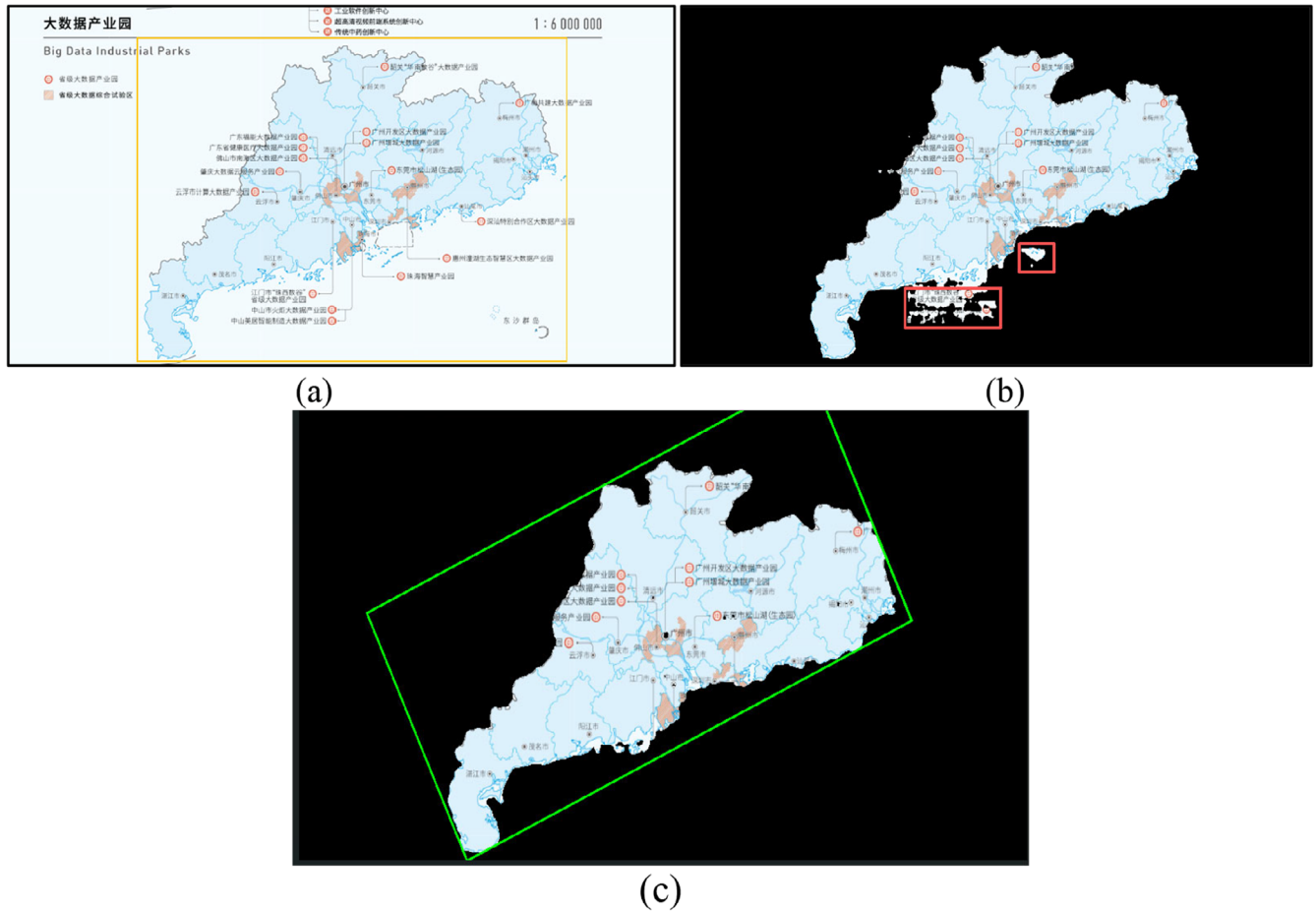
### 4.1 | Layout Classification

In cartographic layout design, not only should the thematic content be considered but also the geometric properties of the base canvas. To systematically analyze the structural patterns of cartographic layouts, we establish a classification scheme for latter analysis based on two fundamental dimensions: page layout and shape of the map area. The two aspects capture both the external constraints of the layout canvas and the intrinsic geometric features of the map area, which are the key factors for map layout design (Yang et al. 2025).

#### 4.1.1 | Page Layout

Page layout, mainly the size and aspect ratio, strongly influence the spatial organization of cartographic elements. Although standard ratios such as 16:9 or 4:3 are commonly used in screen display and printing, actual cartographic design does not necessarily conform to these fixed formats. Therefore, instead of following predefined paper sizes, we classify page layouts purely based on the aspect ratio ( $AR_{\text{page}} = W_{\text{page}} / H_{\text{page}}$ ) of their width ( $W_{\text{page}}$ ) and height ( $H_{\text{page}}$ ) into five categories; the results are shown in Table 2.

As shown in Table 2, we can observe that horizontal layouts dominate the dataset, with horizontal balanced pages (Page-HB) accounting for 38.9% of all maps, followed by horizontal elongated pages (Page-HE) at 7.6%. This indicates a clear preference for horizontally oriented layouts, which may reflect their suitability for accommodating wide map extents and auxiliary elements such as legends and annotations. Vertical layouts are also widely used, with vertical balanced pages (Page-VB) representing 33.3% of the maps, whereas vertical elongated pages (Page-VE) are relatively rare (3.0%). In addition, square layouts (Page-S) constitute 17.2% of the dataset, suggesting that near-square formats remain a common choice for maps with relatively compact spatial extents. Overall, the distribution demonstrates substantial diversity in page layout proportions, highlighting the necessity of aspect-ratio-based classification for capturing real-world cartographic design practices.



**FIGURE 5** | Post-processing of the map area segmentation. (a) Detected bounding box of the map area. (b) Raw segmentation mask of the map area via SAM. (c) Refined segmentation of the map area after post-processing.

**TABLE 2** | Classification of page layout types based on the aspect ratio of the map canvas.

| Categories                     | Definitions                            | Number of maps | Percentage (%) |
|--------------------------------|----------------------------------------|----------------|----------------|
| Horizontal elongated (Page-HE) | $AR_{\text{page}} > 1.5$               | 124            | 7.6            |
| Horizontal balanced (Page-HB)  | $1.2 \leq AR_{\text{page}} \leq 1.5$   | 631            | 38.9           |
| Vertical elongated (Page-VE)   | $AR_{\text{page}} < 0.667$             | 49             | 3.0            |
| Vertical balanced (Page-VB)    | $0.667 \leq AR_{\text{page}} \leq 0.8$ | 541            | 33.3           |
| Square (Page-S)                | $0.8 \leq AR_{\text{page}} < 1.2$      | 280            | 17.2           |
| Total                          |                                        | 1625           | 100            |

#### 4.1.2 | Shape of the Map Area

The shape of the map area also plays a central role in determining layout balance and element placement. Following previous cartographic and graphic design studies (Wei et al. 2024), we characterize the shape using the minimum bounding rectangle (MBR) derived from the segmented outline of the map area. Two key indicators are considered: the aspect ratio of the MBR ( $AR_{\text{box}} = W_{\text{box}}/H_{\text{box}}$ , where  $W_{\text{box}}$  and  $H_{\text{box}}$  denote the width and height), which reflects whether the map is elongated or nearly square; and the orientation of the MBR ( $O_{\text{box}}$ ), which indicates the dominant directional trend of the map body. Based on these indicators, the map area shapes can also be classified into five categories; the results are shown in Table 3.

As shown in Table 3, we can observe that directionally elongated map shapes constitute the majority of the dataset, indicating that most maps area exhibit a clear dominant orientation. Among them, the east–west oriented maps (Map-EW) are the most prevalent, accounting for 31.6% of all samples. This suggests that horizontally extended spatial distributions are common in practical cartographic representations, particularly for regions with wide longitudinal extents. The

**TABLE 3** | Classification of map area shapes based on the aspect ratio and orientation of the minimum bounding rectangle (MBR).

| Categories                      | Definitions                                                                                                                                        | Number of maps | Percentage (%) |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|
| North–South (Map-NS)            | $AR_{\text{box}} > 1.5, AR_{\text{box}} < 0.85, 67.5^\circ \leq O_{\text{box}} \leq 90^\circ$<br>or $-22.5^\circ \leq O_{\text{box}} \leq 0^\circ$ | 138            | 8.5            |
| East–West (Map-EW)              | $AR_{\text{box}} > 1.5, AR_{\text{box}} < 0.85, -22.5^\circ \leq O_{\text{box}} \leq 22.5^\circ$                                                   | 513            | 31.6           |
| Northeast–Southwest (Map-NE–SW) | $AR_{\text{box}} > 1.5, AR_{\text{box}} < 0.85, 22.5^\circ \leq O_{\text{box}} \leq 67.5^\circ$                                                    | 377            | 23.2           |
| Northwest–Southeast (Map-NW–SE) | $AR_{\text{box}} > 1.5, AR_{\text{box}} < 0.85, -67.5^\circ \leq O_{\text{box}} \leq -22.5^\circ$                                                  | 288            | 17.7           |
| Uniform (Map-U)                 | $0.85 \leq AR_{\text{box}} \leq 1.15$                                                                                                              | 309            | 19             |
| Total                           |                                                                                                                                                    | 1625           | 100            |

northeast–southwest (Map-NE–SW) and northwest–southeast (Map-NW–SE) oriented maps account for 23.2% and 17.7% of the samples, respectively, reflecting the frequent occurrence of diagonally oriented spatial patterns in geographic features. In contrast, north–south oriented maps (Map-NS) represent a relatively small proportion of the dataset (8.5%), indicating that vertically elongated spatial distributions are less common. In addition, maps with a relatively uniform shape (Map-U), characterized by near-square bounding rectangles, account for 19.0% of the samples. This suggests that a considerable number of maps depict spatial extents that are balanced in both horizontal and vertical directions. Overall, the observed distribution highlights the diversity of map area shape characteristics and underscores the importance of incorporating both aspect ratio and orientation in the classification of cartographic layouts.

## 4.2 | Layout Evaluation Metric Metrics for Evaluating Layout Design Effectiveness

To evaluate map layout quality, researchers have proposed a variety of measures that provide insights into how layouts support visual communication and usability. Building on these ideas, our goal is to establish a systematic evaluation framework for the collected maps. A well-designed layout should balance the visual prominence of the map area, the accessibility of auxiliary elements, and the overall efficiency of information delivery. To this end, we draw on principles from multiple fields, including cartography, visual design, and human-computer interaction, and collect a set of metrics that capture the qualities of map layouts, as follows:

- **Hierarchy:** The layout should organize the map area and auxiliary elements into a coherent visual hierarchy, ensuring that visual weights are appropriately distributed according to their functional roles in the thematic narrative.
- **Visual balance:** Map elements should be placed in visually stable positions, avoiding excessive offset from the perceptual center.
- **Compactness:** The page space should be used efficiently, minimizing unused areas and ensuring appropriate density of elements.
- **Alignment:** Elements should follow consistent positional logic, aligning with page geometry or the map area orientation.

For each principle, we have identified corresponding measures that can be directly computed from the annotated dataset and are summarized in Table 4, which provides the basis for the subsequent statistical analysis of layout patterns.

## 5 | Statistical Analysis and Results

### 5.1 | Descriptive Analysis

#### 5.1.1 | Distribution of Evaluation Metric

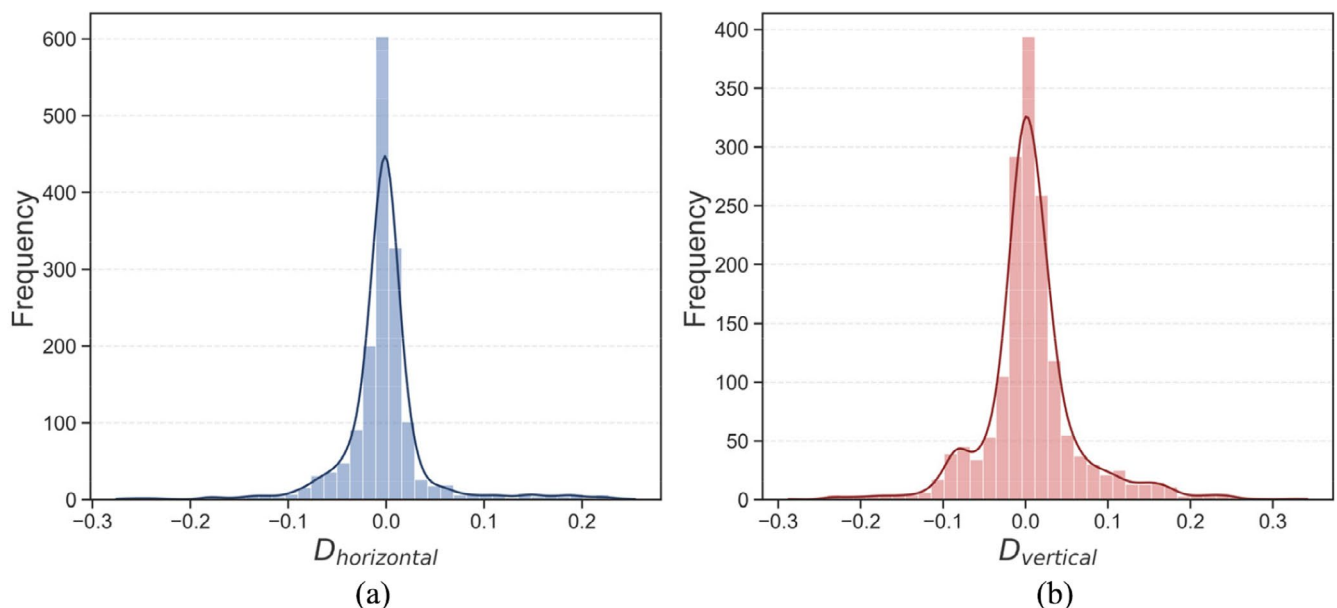
To quantitatively analyze the structural characteristics of map layouts, we evaluated the dataset based on the proposed four design principles: hierarchy, balance, compactness, and alignment. Each principle is operationalized by specific indicators (Table 4).

**5.1.1.1 | Hierarchy.** According to Table 4, we evaluated the hierarchy of the layout by calculating the normalized offsets ( $D_{\text{horizontal}}$  and  $D_{\text{vertical}}$ ) of the main map's area centroid relative to the page center. The frequency distributions of these offsets are presented in Figure 6. As shown in the figure, we can observe that both horizontal and vertical offsets exhibit a highly concentrated unimodal distribution around zero. Quantitatively, the vast majority of main maps are positioned within a narrow range (e.g., deviation  $< 10\%$ ) of the geometric center of the page. This statistical pattern provides strong empirical evidence for the principle of centrality, confirming that placing the thematic subject at the center is the dominant strategy to establish visual hierarchy and ensure user attention. Furthermore, the distribution tail is short and extreme deviations are rare, indicating that designers systematically avoid placing the map area at the page margins. Interestingly, the presence of minor deviations within the central peak suggests that while centrality is the goal, strict geometric alignment is not rigidly enforced. Instead, designers likely adjust the map's position to align with the visual center (optical center) rather than the absolute geometric center, thereby achieving a more natural and stable visual perception.

**5.1.1.2 | Visual Balance.** According to Table 4, we applied  $B_{\text{horizontal}}$  and  $B_{\text{vertical}}$  to evaluate the visual balance of the map in horizontal and vertical directions, and the distributions are illustrated with histograms in Figure 7. As shown in the figure, both distributions exhibit a distinct positive skewness (or long-tail

**TABLE 4** | Quantitative evaluation metrics for map layout quality.

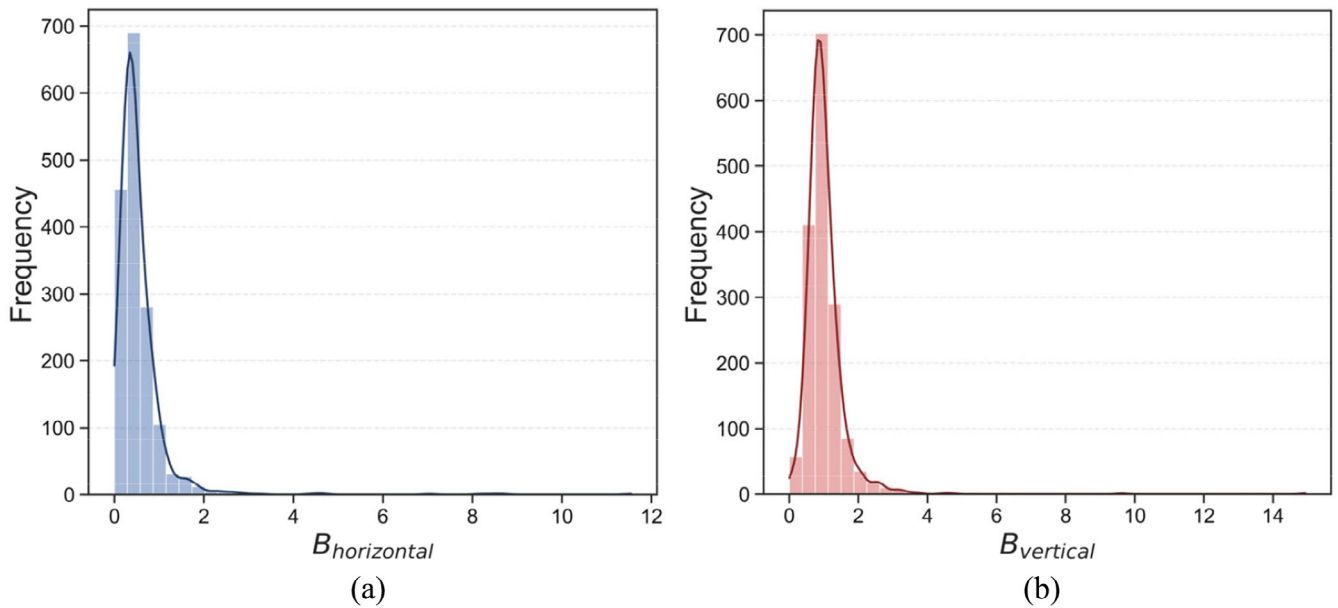
| Principles     | Measure                                                   | Description and definition                                                                                                                                                                                                                                                                                                                                                                                                                                       | Source                  |
|----------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Hierarchy      | $D$ ( $D_{\text{horizontal}}$ and $D_{\text{vertical}}$ ) | Evaluated by the distance of the main map centroid from the visual center, distinguishing horizontal offset ( $D_{\text{horizontal}}$ ) and vertical offset ( $D_{\text{vertical}}$ ).                                                                                                                                                                                                                                                                           | Dent et al. (2008)      |
| Visual balance | $B$ ( $B_{\text{horizontal}}$ and $B_{\text{vertical}}$ ) | Evaluated using weighted contributions of map elements, considering both horizontal (left-right) and vertical (top-bottom) balance to ensure layout stability. Thus, it has two visual balance in top-bottom ( $B_{\text{horizontal}}$ ) and left-right ( $B_{\text{vertical}}$ )                                                                                                                                                                                | O'Donovan et al. (2014) |
| Compactness    | $N$<br>$R_{\text{map}}$                                   | Number of map elements.<br>$R_{\text{map}} = A_{\text{map}}/A_{\text{page}}$ , the proportion of the map area ( $A_{\text{map}}$ ) to page area ( $A_{\text{page}}$ ).                                                                                                                                                                                                                                                                                           | Wei et al. (2017)       |
| Alignment      | $R_{\text{horizontal}}$<br>$R_{\text{vertical}}$          | $R_{\text{horizontal}} = C_{\text{horizontal}}/L_{\text{horizontal}}$ , where $C_{\text{horizontal}}$ is the number of horizontally aligned sight lines, and $L_{\text{horizontal}}$ is the total number of horizontal sight lines.<br>$R_{\text{vertical}} = C_{\text{vertical}}/L_{\text{vertical}}$ , where $C_{\text{vertical}}$ is the number of horizontally aligned sight lines, and $L_{\text{vertical}}$ is the total number of horizontal sight lines. | O'Donovan et al. (2014) |

**FIGURE 6** | Frequency distributions of the normalized layout offsets of the map area relative to the page center. (a) Histogram of the horizontal offset ratio ( $D_{\text{horizontal}}$ ); (b) Histogram of the vertical offset ratio ( $D_{\text{vertical}}$ ). The x-axis represents the deviation as a proportion of the page dimensions, where 0 indicates the geometric center.

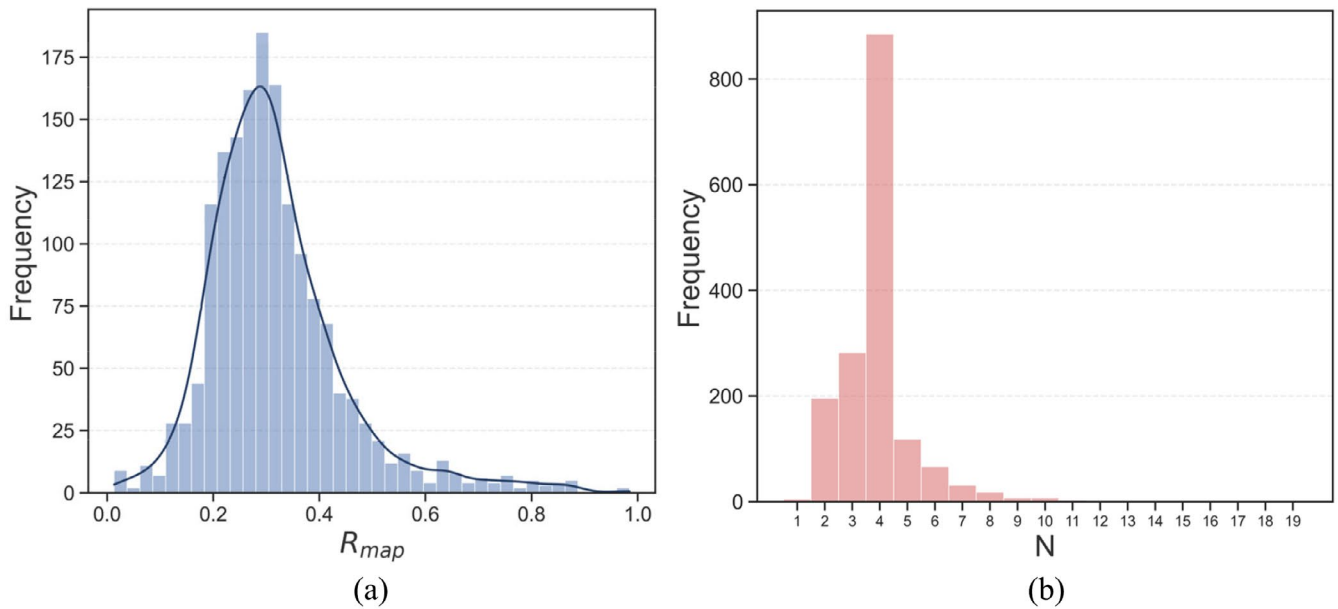
distribution). The vast majority of maps cluster tightly within the low-value range (0–2), with frequencies decaying rapidly as the balance index increases. This pattern indicates that while maximizing visual balance is a core design objective, strict mathematical symmetry is not rigidly enforced. Instead, designers adopt a pragmatic strategy, maintaining visual equilibrium within a reasonable tolerance range to accommodate the inherent irregularity of cartographic elements. This approach ensures a stable compositional structure that enhances legibility without imposing artificial rigidity.

**5.1.1.3 | Compactness.** According to Table 4, we evaluated layout compactness through the total number of map elements ( $N$ ) and the area utilization ratio of the map area ( $R_{\text{map}}$ ). The results of these metrics are illustrated in Figure 8 and Table 5.

As shown in Figure 8a, the distribution of  $R_{\text{map}}$  exhibits a positive skew, with values predominantly concentrated between 0.2 and 0.4 and peaking at 0.3. This indicates that the map area typically occupies approximately one-third of the page area. This



**FIGURE 7** | Frequency distributions of the visual balance metrics. (a) Histogram of the horizontal (left–right) balance degree ( $B_{\text{horizontal}}$ ); (b) Histogram of the vertical (top–bottom) balance degree ( $B_{\text{vertical}}$ ). Note that lower values indicate a higher degree of visual equilibrium.



**FIGURE 8** | Distribution of compactness metrics. (a) Histogram of the map area ratio ( $R_{\text{map}}$ ); (b) Histogram of the number of map elements ( $N$ ).

ratio demonstrates a sophisticated balance: it ensures the visual dominance of the thematic subject while reserving sufficient “breathing room” (the remaining 60%–80% of the page) for the title, legend, and other supporting metadata. The rarity of extremely high utilization ratios ( $R_{\text{map}} > 0.7$ ) further suggests that designers systematically avoid “crowding out” auxiliary information, ensuring that the map remains a functional document rather than a purely aesthetic image.

As shown in Figure 8b, the element counts range from 0 to 19, with a high concentration around 4 elements, reaching a peak frequency of approximately 1200 samples. This distribution suggests that designers generally adhere to a moderate density strategy, prioritizing clarity by avoiding both overly sparse and cluttered

layouts. The peak at  $N=4$  aligns with the “core components” identified in Table 5, typically comprising the Map Area (100%), Title (91.14%), Legend (88.86%), and Scale (65.85%). The high prevalence of these four elements confirms their status as the functional pillars of thematic cartography. However, the fact that the Title and Legend do not reach 100% indicates that in professional series (e.g., specific plates within an atlas), these elements may be moved to page headers or omitted if the symbology is self-explanatory. Conversely, auxiliary elements exhibit significantly lower frequencies (Table 5), such as Description (15.63%), Statistics (11.94%), and Inset map (10.58%). Notably, the North Arrow appears in only 1.85% of the maps. This underscores the context-dependent necessity of orientation indicators; in many contemporary thematic maps, the North Arrow is considered nonessential if a standard

orientation is assumed or if the geographic context is globally recognized, thereby preserving “white space” for better legibility.

**5.1.1.4 | Alignment.** According to Table 4, we employed the horizontal ( $R_{\text{horizontal}}$ ) and vertical ( $R_{\text{vertical}}$ ) alignment rates to evaluate the structural discipline of the map layouts. The frequency distributions of these metrics are illustrated in Figure 9.

As shown in the figure, both horizontal and vertical alignment rates are predominantly concentrated within the 0–0.4 range, with peak frequencies observed at approximately 500 for horizontal alignment and 600 for vertical alignment. Higher alignment rates (e.g., >0.6) are notably infrequent. This statistical evidence suggests that strict geometric alignment is not the dominant practice in professional cartography. Instead, the data

**TABLE 5** | Frequency statistics of detected cartographic elements in the dataset.

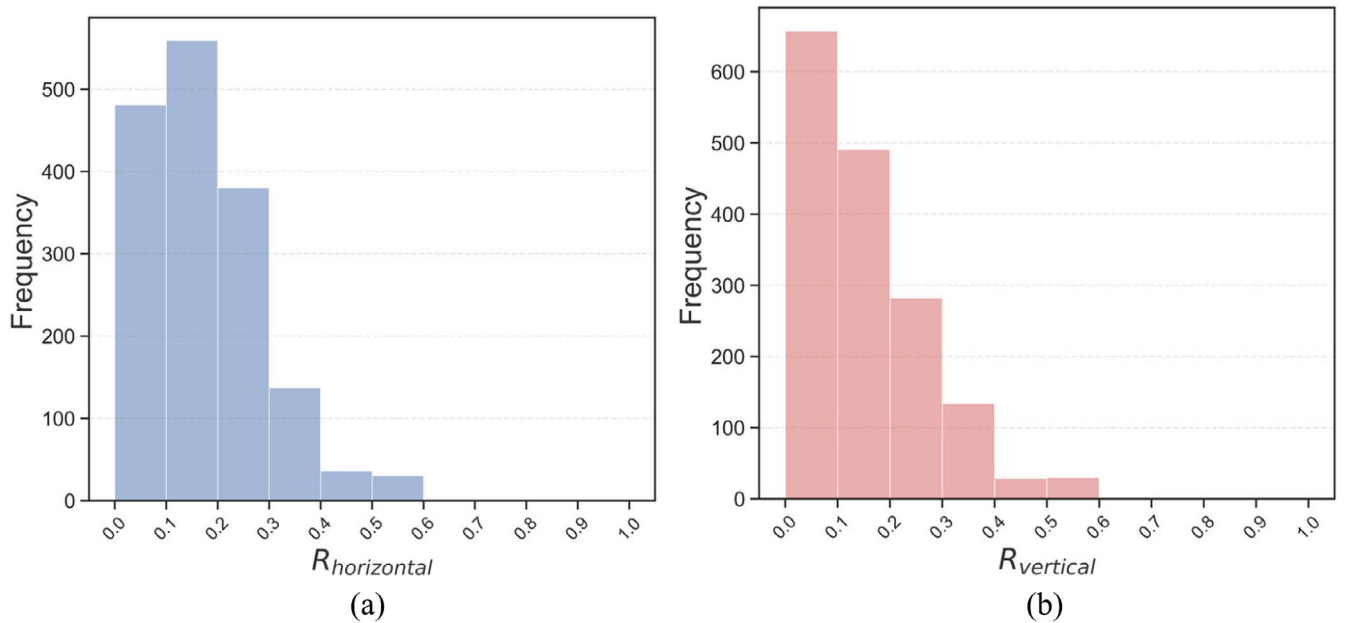
| Element Type | Total Count | Frequency (%) |
|--------------|-------------|---------------|
| Map Area     | 1625        | 100.00%       |
| Title        | 1481        | 91.14%        |
| Legend       | 1444        | 88.86%        |
| Scale        | 1070        | 65.85%        |
| Description  | 254         | 15.63%        |
| Statistics   | 194         | 11.94%        |
| Inset map    | 172         | 10.58%        |
| Picture      | 91          | 5.60%         |
| Table        | 40          | 2.46%         |
| North Arrow  | 30          | 1.85%         |

reveals a pragmatic alignment strategy that balances structural order with compositional flexibility.

While traditional graphic design often emphasizes a rigid grid system to ensure consistency, map layout design must accommodate the irregular and organic shapes of geographic features. The moderate alignment levels (0.2–0.4) observed in the dataset indicate that designers maintain a “sufficient” degree of alignment to establish a clear visual hierarchy and reading path, while intentionally introducing variations to avoid a rigid or monotonous aesthetic. This selective alignment approach allows for a more dynamic composition, where auxiliary elements are positioned to complement the map area’s unique contours rather than being forced into a restrictive geometric template. Consequently, the layout achieves a harmonious trade-off between structural clarity and creative adaptability, ensuring that the map remains both functional and visually engaging.

**5.1.1.5 | Summary.** Overall, the descriptive analysis demonstrates that the layout patterns of exemplary thematic maps largely converge with established cartographic design conventions, yet also reveal significant practical adaptations. The high degree of centrality observed in the hierarchy analysis and the consistently low values in visual balance confirm that visual stability and thematic dominance remain the primary structural objectives. Compactness is maintained through a disciplined selection of elements, typically centered around a “core four” (Map Area, Title, Legend, and Scale), ensuring that the map area retains a reasonable area ratio ( $R_{\text{map}}$ ) to balance prominence with legibility.

However, the empirical data on alignment metrics ( $R_{\text{horizontal}}$ ,  $R_{\text{vertical}}$ ) offers a nuanced departure from rigid graphic design theories. Instead of pursuing maximum geometric alignment, professional practice favors moderate alignment coupled with compositional flexibility. This suggests that in the complex context of cartography—where irregular geographic shapes must



**FIGURE 9** | Distribution of alignment metrics. (a) Histogram of the horizontal alignment degree ( $R_{\text{horizontal}}$ ); (b) Histogram of the vertical alignment degree ( $R_{\text{vertical}}$ ).

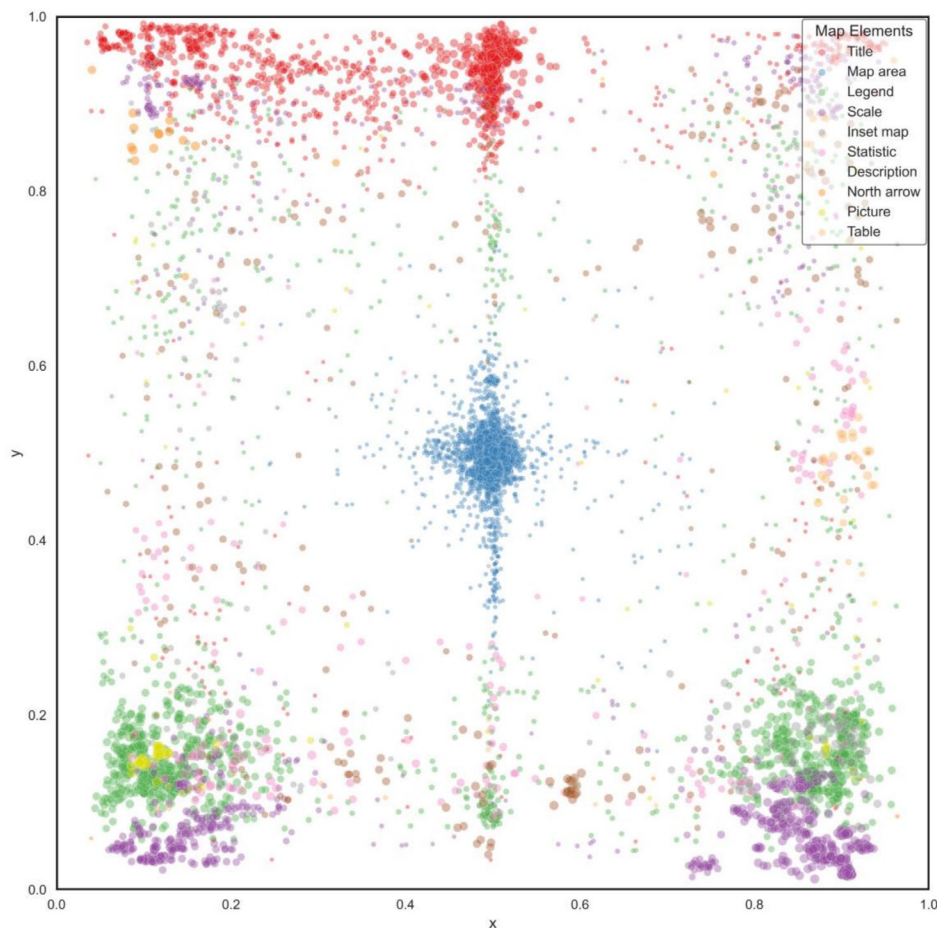
be integrated with structured metadata—designers prioritize a pragmatic balance over strict grid-based adherence. Such a divergence underscores the value of large-scale empirical analysis: it captures the “tacit knowledge” of professional designers, proving that effective layouts are governed by a “Center-oriented but Flexible” philosophy that ensures both structural integrity and creative adaptability.

### 5.1.2 | Distribution of Locations of Map Elements

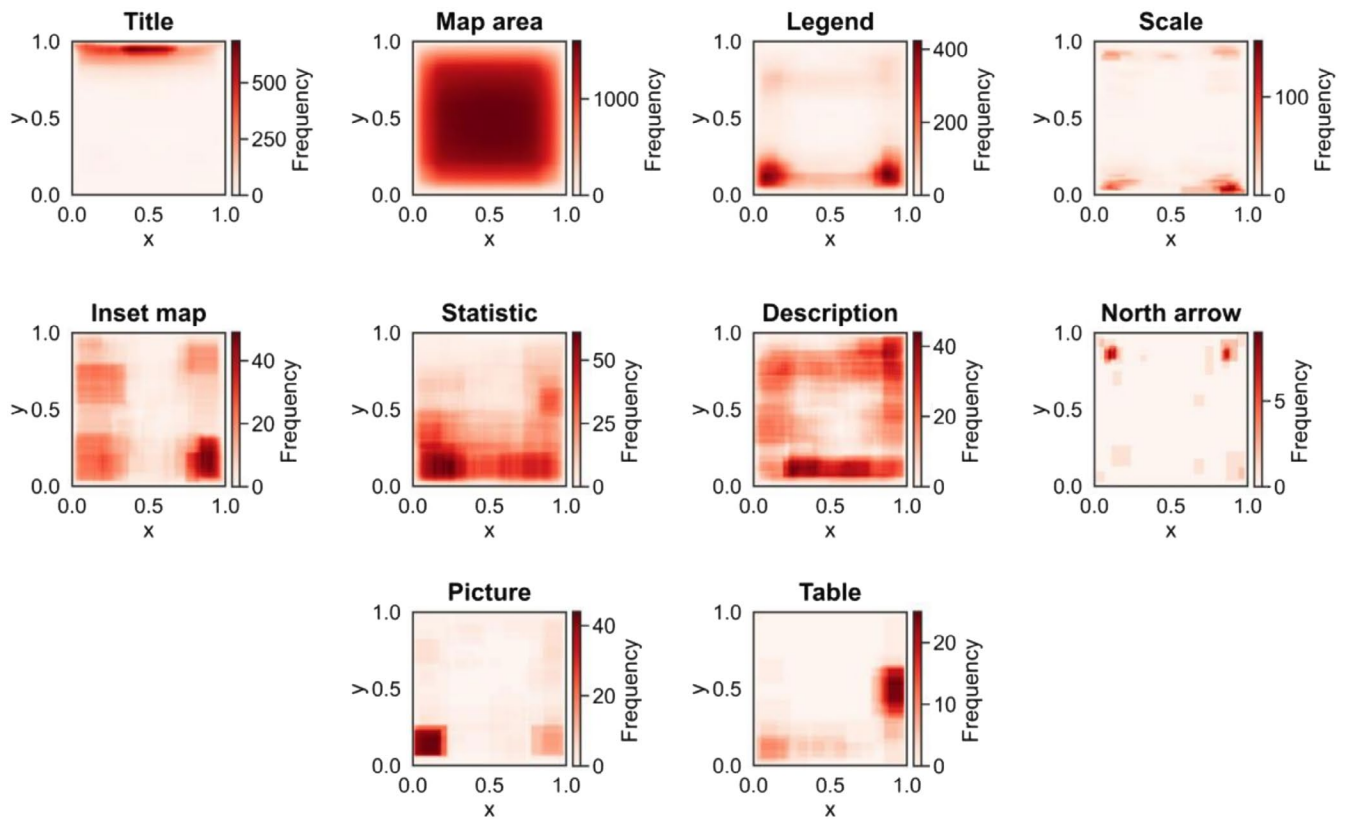
To analyze the spatial distribution of map elements, we normalized the centroid coordinates of each detected element to the page coordinate system and then visualized them using a point-density approach. The result is shown as Figure 10. However, because map elements possess distinct shapes and varying dimensions, relying solely on centroids may oversimplify the composition and fail to capture the true footprint of each element. Therefore, to complement the centroid analysis, we additionally generated spatial area heatmaps (Figure 11). These heatmaps accumulate the full bounding box areas of the elements, illustrating their actual spatial coverage and physical occupancy on the canvas. Together, Figures 10 and 11 provide two complementary perspectives: Figure 10 reveals where designers tend to place elements, while Figure 11 shows how much space those elements actually occupy.

From the figure, we can observe that the map title elements are highly concentrated in the upper-left and upper-central regions of the page, forming distinct clusters. This conforms to conventional design practice, where titles are placed at the top to guide the initial reading direction. The map area forms a strong central concentration in Figure 10, while Figure 11 further shows that it occupies most of the central page area. This pattern confirms the dominant and compact role of the map area, ensuring strong visual prominence of thematic content. In the lower part of the layout, the legend and scale bar are mainly distributed in the lower-left and lower-right corners. Although slightly more concentrated on the left, the two show an overall symmetric placement pattern, contributing to visual stability and compositional balance. Other elements such as descriptions, inset pictures, north arrows, and tables exhibit relatively scattered distributions. They are typically positioned along page margins or in residual blank areas, making efficient use of available space while avoiding interference with the core map content.

In summary, the spatial distribution of elements demonstrates that layouts are organized around a visually dominant map area, with supporting elements positioned to enhance readability and stability, while peripheral components are placed flexibly at the margins to maximize space efficiency and preserve clarity. These findings highlight a coherent design logic that balances functionality and aesthetics.



**FIGURE 10** | Distribution of locations of map elements. The horizontal and vertical axes represent the normalized spatial coordinates (X and Y) of the map canvas, scaled from 0 to 1.



**FIGURE 11** | Area-based spatial distribution heatmaps for 10 categories of map elements.

## 5.2 | Correlation and Significance Tests

To further uncover the correlations in map layouts, we conducted correlation analysis to examine whether significant associations exist between categorical variables (e.g., page layout types and map shapes) and whether monotonic relationships emerge among quantitative metrics for evaluating layout design effectiveness. Specifically, Pearson's chi-squared test was employed for analysis on correlation between page layouts types and map shapes, as it is well suited for detecting nonrandom dependencies between classification variables (Pearson 1900). Meanwhile, Spearman's rank correlation was applied to indicator values, which does not assume normality and is robust against outliers (Spearman 1987), thus appropriate for analyzing monotonic relationships among layout metrics.

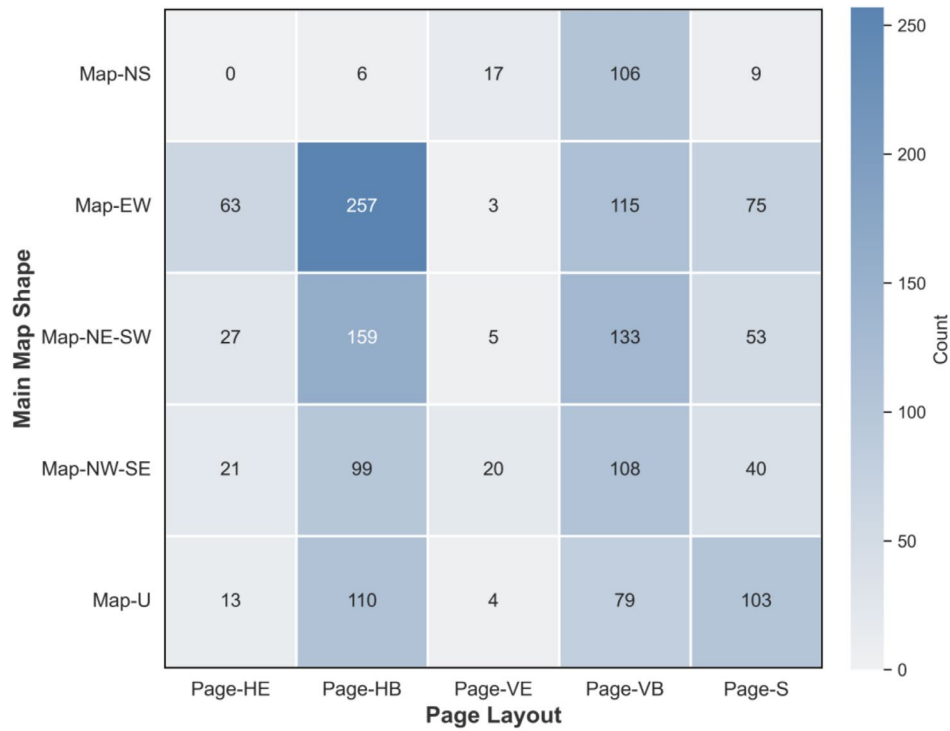
### 5.2.1 | Correlation Between Page Layout Types and Map Area Shapes

The Pearson's chi-squared test was used to assess the association between map shape and page layout type. Results demonstrate a highly significant relationship ( $\chi^2 = 329.422$ ,  $p < 0.001$ ), indicating that the two categorical variables are not independent. This strong correlation is visually explicated by the cross-tabulation heatmap (Figure 12), which exhibits distinct clustering patterns based on geometric similarity. As shown in the figure, the row corresponding to North–South oriented maps presents a clear “hot spot” in the Vertical-balanced and Vertical-elongated

columns, whereas the cells for Horizontal-elongated layouts are virtually empty. This indicates that vertical geography almost exclusively dictates a vertical page format. Conversely, regarding East–West oriented maps, the highest frequency density is concentrated within the Horizontal-balanced and Horizontal-elongated categories. This divergence underscores a strict dependency between the “content” and the “container”: designers systematically select a page aspect ratio that mirrors the directional elongation of the map area to minimize invalid white space and maximize the display area.

### 5.2.2 | Correlation Among Metrics for Evaluating Layout Design Effectiveness

We further applied Spearman's correlation analysis to explore interrelationships among layout indicators, as shown in Figure 13. Several significant monotonic relationships were identified. The horizontal offset of the map area centroid shows a moderate negative correlation with left–right balance ( $r = -0.54$ ,  $p < 0.001$ ), while vertical offset exhibits a similar negative correlation with top–bottom balance ( $r = -0.496$ ,  $p < 0.001$ ), confirming that deviations of the map area from the visual center directly undermine layout stability. In addition, the number of elements is positively correlated with both horizontal ( $r = 0.437$ ,  $p < 0.001$ ) and vertical alignment rates ( $r = 0.222$ ,  $p < 0.001$ ), suggesting that as element counts increase, designers tend to impose stronger alignment to maintain readability.



**FIGURE 12** | Correlation between page layout types and main map area shapes.

### 5.3 | Variance Analysis

Page layout and map area shape are two core factors that fundamentally determine the map layout. To better understand how these factors affect layout performance, we tested their influences on the proposed evaluation metrics. As the metrics are not normally distributed and the homogeneity of variance assumption is not guaranteed across groups. Under these conditions, traditional parametric ANOVA is not appropriate. To address this, we employed the Scheirer-Ray-Hare test, a rank-based nonparametric method for two-way variance analysis (Scheirer et al. 1976). This method allows us to simultaneously examine the main effects of both page layout and map area shape as well as their interaction effects; the results are shown in Table 6. From Table 6, we can observe that both page layout type and map area shape exert significant influences on most metrics, with their interaction also statistically significant in many cases. The only exceptions are that vertical alignment does not vary significantly across page layouts, and horizontal alignment does not differ significantly across map area shapes. These results suggest that layout outcomes are shaped not only by each factor independently but also by their combined influence.

To further clarify the sources of interaction, we performed a simple effect analysis, in which the main map shape was fixed at each category and the differences among page layout types were examined. The results are shown in Table 7. As shown in the table, we can observe that the influence of page layout is highly shape-dependent.

For east–west oriented main maps (EW), significant differences are primarily observed in map element count ( $N$ ) and vertical

offset ( $D_{\text{vertical}}$ ), where horizontal page layouts—particularly horizontal-balanced (HB) and horizontal-elongated (HE)—consistently outperform square or vertical layouts. Beyond these two metrics, Table 7 also reports significant between-layout differences in  $D_{\text{horizontal}}$ ,  $D_{\text{vertical}}$ ,  $R_{\text{map}}$ , and  $R_{\text{horizontal}}$ , indicating that EW maps are simultaneously sensitive to centroid positioning, space utilization, and alignment structure. This indicates that horizontally extended geographic extents benefit from page formats that align with their dominant direction, thereby improving compactness and positional stability. In contrast, diagonally oriented main maps (NE–SW and NW–SE) exhibit more complex sensitivity patterns. Significant differences emerge across multiple metrics, including visual balance ( $B_{\text{vertical}}$ ) and centroid offsets ( $D_{\text{vertical}}$ ) and element count ( $N$ ), but no single page layout type dominates across all measures. This suggests that layouts for diagonally oriented maps require more nuanced trade-offs between page geometry and map orientation, rather than reliance on a single optimal template. For uniformly shaped main maps (Map-U), square pages (Page-S) and horizontal layouts demonstrate significant advantages across several metrics, particularly in centroid offsets ( $D_{\text{vertical}}$ ) and element count ( $N$ ). This finding indicates that when the main map area lacks a strong directional bias, the geometric proportion of the page itself becomes a primary factor governing layout quality.

In summary, page layout and main map shape exert substantial influence on cartographic layout design. However, their impacts are not uniform: different orientations of the main map area emphasize different evaluation metrics, such as offsets, utilization, balance, or element count. This suggests that effective map design should not rely on a one-size-fits-all approach but instead consider the interplay between page geometry and map



**FIGURE 13** | Correlation among metrics for evaluating layout design effectiveness.

**TABLE 6** | Results of the Scheirer-Ray-Hare non-parametric two-way variance analysis. The table reports the main effects of page layout type and main map shape, as well as their interaction effects, on each metric for evaluating layout design effectiveness.

| Measure          | Page layout  | Main map shape | Interaction effect |
|------------------|--------------|----------------|--------------------|
| $D_{horizontal}$ | *** (<0.001) | *** (<0.001)   | *** (<0.001)       |
| $D_{vertical}$   | *** (<0.001) | *** (<0.001)   | *** (<0.001)       |
| $B_{horizontal}$ | 0.099        | *** (<0.001)   | ** (0.002)         |
| $B_{vertical}$   | *** (<0.001) | *** (<0.001)   | *** (<0.001)       |
| $N$              | *** (<0.001) | *** (<0.001)   | *** (<0.001)       |
| $R_{map}$        | *** (<0.001) | *** (<0.001)   | *** (<0.001)       |
| $R_{horizontal}$ | *** (<0.001) | 0.104          | *** (<0.001)       |
| $R_{vertical}$   | 0.219        | *** (<0.001)   | ** (0.003)         |

**TABLE 7** | Pairwise comparisons between layout metrics (see Table 4) and different main map shapes (EW, NS, NE–SW, NW–SE, U, see Table 3) across varying page layout types (HE, HB, VE, VB and S, see Table 2). Statistical significance is denoted with asterisks (\*, \*\*, \*\*\*) corresponding to *p*-values. Only rows and columns with statistically significant correlations have been retained in this table.

| Metric                  | EW                                                                                        | NE–SW                                                                                                          | NW–SE                                                                    | U                                                                                                                                 |
|-------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| $B_{\text{horizontal}}$ | —                                                                                         | HB/VB (0.197**)                                                                                                | —                                                                        | —                                                                                                                                 |
| $B_{\text{vertical}}$   | S/HB (0.366**)<br>S/VE (4.172*)<br>VB/VE (4.266*)<br>HB/VE (4.537***)<br>VE/HE (4.576***) | S/HB (0.296***)<br>S/VB (0.262**)<br>HB/HE (0.150*)                                                            | HB/VB (0.157)<br>HB/VE (0.252**)                                         | VB/S (0.131**)<br>VB/HB (0.160*)<br>S/HB (0.291***)<br>S/HE (0.305**)                                                             |
| $D_{\text{horizontal}}$ | VB/HE (0.026*)                                                                            | HB/VB (0.002***)                                                                                               | —                                                                        | VB/HB (0.021***)<br>VB/HE (0.050***)<br>S/HB (0.014***)<br>S/HE (0.043**)                                                         |
| $D_{\text{vertical}}$   | VB/HB (0.061***)<br>VB/HE (0.055***)                                                      | HE/VE (0.044*)                                                                                                 | HB/VE (0.042***)<br>S/VE (0.037**)<br>VB/VE (0.034**)<br>VE/HE (0.028**) | VB/S (0.047***)<br>S/HB (0.049***)                                                                                                |
| $N$                     | S/HB (1.290***)<br>VB/HB (1.214***)<br>HB/HE (0.873***)                                   | S/HB (1.547***)<br>S/HE (1.256***)<br>S/VB (0.811***)<br>HB/VB (0.736***)<br>HB/VE (1.898*)<br>VB/VE (1.162**) | HB/S (0.769**)<br>HB/VB (0.505**)                                        | VB/S (1.213***)<br>VB/HB (0.645***)<br>VB/HE (1.665***)<br>S/HB (1.858***)<br>S/VE (1.840*)<br>S/HE (2.878***)<br>HB/HE (1.020**) |
| $R_{\text{map}}$        | S/HE (0.072*)<br>VB/HB (0.111***)<br>VB/HE (0.129***)                                     | —                                                                                                              | HB/HE (0.085**)<br>S/HE (0.086*)<br>VB/HE (0.092***)                     | VB/S (0.094***)<br>S/HB (0.129***)                                                                                                |
| $R_{\text{horizontal}}$ | S/HB (0.065**)<br>S/HE (0.083*)<br>VB/HB (0.060**)<br>VB/HE (0.078*)                      | S/HB (0.074*)<br>S/HE (0.128**)<br>HB/VB (0.083***)<br>HE/VB (0.137***)<br>HE/VE (0.204*)                      | HB/VE (0.087*)<br>S/VE (0.116*)                                          | VB/HB (0.097***)<br>S/HB (0.115***)<br>S/HE (0.162**)                                                                             |
| $R_{\text{vertical}}$   | —                                                                                         | —                                                                                                              | —                                                                        | S/HE (0.103*)<br>HB/HE (0.099*)                                                                                                   |

orientation. Such differentiated patterns provide valuable insights for developing layout recommendation strategies tailored to specific main map shapes.

## 6 | Discussion

### 6.1 | Potential Usage

With a large dataset of high-quality maps, clear classification schemes, and quantitative evaluation metrics, it may support both the exploration of layout patterns and the improvement of design practice in the future.

#### 6.1.1 | Layout Exploring

This study has provided an annotated dataset of exemplary maps, which systematically distinguishes between page layouts,

main map shapes, and a range of quantitative evaluation metrics. Thus, it can offer users and researchers a valuable resource for exploring the diversity of cartographic layouts. By comparing different categories of page aspect ratios, main map orientations, or balance and alignment measures, users can gain a deeper understanding of how design choices affect both the structure and aesthetics of a map. In this sense, the dataset could serve as a reference for best practices.

#### 6.1.2 | Inspiring Layout Designs and Evaluation

The framework developed in this study offers practical support for layout design through both evaluation and recommendation. Because the system can automatically recognize cartographic elements and calculate layout metrics, users are able to obtain objective feedback on the strengths and weaknesses of their own designs. For example, a student may upload their map layout to receive an evaluation of element balance or alignment,

highlighting areas where refinement is needed. In addition, the framework can serve as a generative tool for recommending new designs due to derived layout rules. Users may upload a main map and specify preferences such as page layout type or desired number of supplementary elements. Based on the patterns revealed in our dataset, the system can then suggest layout alternatives—for example, positioning the legend and scale symmetrically at the bottom corners, or proposing a vertically balanced composition when the main map area is oriented north–south. For casual or non-expert users, it could lower the barrier to map creation by automatically generating aesthetically pleasing layouts.

## 6.2 | Design Flexibility vs. Rigid Rules

Our empirical results suggest that while many observed patterns confirm long-standing cartographic wisdom, design practice also exhibits substantial flexibility. For example, the concentration of titles at the upper positions and the centrality of the main map area are consistent with conventional expectations. Likewise, the tendency of legends and scales to cluster near the lower corners aligns with traditional guidance regarding balance and stability. These results confirm the persistence of established design logics across diverse map sources.

However, at the same time, not all results adhere to rigid rules. The distribution of alignment metrics shows that strict geometric alignment is relatively uncommon; instead, moderate alignment dominates. This observation highlights the creative leeway exercised by designers, who often choose to balance structural order with aesthetic variation. Rather than enforcing mechanical symmetry, maps frequently employ subtle misalignments that produce dynamism without compromising readability. This finding underscores the importance of acknowledging flexibility as an integral component of cartographic design, suggesting that automated systems should avoid overly rigid optimization and instead incorporate controlled variability to mimic authentic design practices. In real design practice, flexibility can also empower users. Professional cartographers may want to deviate from rules for artistic or thematic emphasis, while novice users may benefit from rule-based guidance. Thus, future systems could offer dual modes: one strict (for novices or formal outputs) and one flexible (for experts or creative design).

## 6.3 | Limitation and Future Work

Although this study systematically constructed a large dataset of exemplary maps, developed quantitative metrics, and revealed generalizable patterns of cartographic layouts, several limitations remain that also open avenues for future work.

First, the dataset is limited in scope and primarily composed of professional cartographic works. While these maps provide reliable exemplars for extracting layout rules, they may not fully capture the diversity of real-world cartographic practices, especially in contexts where design conventions are less strictly followed. Future research should expand the dataset to include a broader range of maps, such as educational,

community-based, or crowd-sourced productions, in order to validate whether the observed patterns hold across different domains.

Second, most maps analyzed in this study adhere to relatively standard cartographic forms. However, the recent emergence of highly artistic and experimental map designs challenges these conventions and introduces novel visual forms that are difficult to capture with current recognition methods. Our element detection and segmentation pipeline demonstrates high accuracy on traditional maps but is less effective when applied to these artistic cases. Improving recognition models to handle artistic variation, such as hand-drawn, stylized, or mixed-media maps, represents an important future challenge.

Third, this research focused primarily on the layout dimension of cartographic design and relied on quantitative metrics derived from detected element positions and shapes. These metrics provide a useful way to analyze layout design effectiveness from the perspectives of hierarchy, balance, compactness, and alignment. However, map design quality is a complex perceptual and design concept that involves not only layout geometry but also the integration among elements, semantic relationships, aesthetic coherence, color, typography, symbolization, and the user's reading process. These holistic factors cannot be fully captured by the current metric set. Future work could incorporate additional design dimensions, perceptual experiments, expert evaluations, and user interaction data to validate and extend these metrics toward a more comprehensive assessment of cartographic quality.

Finally, the present study emphasizes the identification and quantification of layout patterns, but it remains primarily descriptive in nature. To translate these insights into practical value, future efforts should focus on developing application-oriented systems, such as intelligent layout recommendation tools or semi-automated cartographic design assistants, which have been discussed in *Section 6.1*. Such systems could embed the identified rules and metrics into interactive workflows, providing designers with actionable guidance while also preserving flexibility for creative adaptation, which would empower ordinary users to participate more actively in map creation. Moreover, the interactive process itself could serve as a valuable channel for collecting new maps and user-generated layouts, thereby enriching both the dataset and the system's adaptability over time.

## 7 | Conclusion

This study presents a data-driven investigation of cartographic layout design based on a curated dataset of 1625 exemplary thematic maps. Unlike existing generic datasets that passively aggregate ubiquitous maps, our dataset was rigorously distilled from over 20,000 raw images to ensure it represents good cartographic design excellence. By combining automated cartographic element recognition with quantitative evaluation metrics, we move beyond anecdotal design experience and provide large-scale empirical evidence on how professional map layouts are structured in practice. The analysis confirms several long-standing cartographic conventions, including the central

placement of the main map area, a strong tendency toward visually balanced layouts, and the stable positioning of core elements such as titles and legends. At the same time, the results reveal clear departures from traditional assumptions. Notably, elements commonly regarded as standard—such as the north arrow—appear in only a small fraction of exemplary maps (1.85%). In addition, strict geometric alignment is not dominant; instead, designers favor moderate and flexible alignment strategies that adapt to irregular map shapes while maintaining readability. Importantly, the findings show that layout effectiveness is strongly shape-dependent. No single page layout consistently performs best across different main map shapes, indicating that cartographic design cannot be reduced to universal templates. Ultimately, the core contributions of this work are three-fold. Methodologically, we developed a robust deep-learning pipeline to automatically recognize and extract complex layout structures from unstructured map images. Theoretically, we shifted the focus from mere object detection to the scientific measurement of layout patterns, computationally abstracting implicit cartographic design rules into explicit, reproducible knowledge. Practically, the extracted design knowledge provides empirical references and actionable layout recommendations for automated or semi-automated map design systems. By operationalizing hierarchy, balance, compactness, and alignment into computable metrics, this study provides an empirical foundation for intelligent map layout analysis and supports the development of data-driven evaluation and recommendation methods in future cartographic design systems. Future work will extend the framework to incorporate additional design dimensions such as color and typography, and translate the analytical insights into interactive or semiautomatic layout design tools that support human–AI collaboration.

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## Disclosure

The authors have nothing to report.

## Conflicts of Interest

The authors declare no conflicts of interest.

## Data Availability Statement

The datasets generated and analyzed during the current study are not publicly available at this stage. This is due to ongoing analyses and planned extensions of the dataset. However, the data are available from the corresponding author upon reasonable request.

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